

Solution Architecture & ER Model

1. MVC Pattern Design

- Model: Decides flood class using thresholds.
- View: Frontend templates with dark Navy design.
- Controller: Flask routes and error handlers.

2. Entity-Relationship (ER) Catalog

The system architecture tracks four primary logical entities:

- USER: user_id, username, email, role.
- WEATHER_DATA: data_id, user_id, cloud_cover, annual, jan_feb, mar_may, jun_sep.
- MODEL: model_id, model_name, algorithm_type, accuracy.
- PREDICTION: prediction_id, data_id, model_id, prediction_result, confidence, prediction_date.